Ratio and Proportion - Mark scheme

Number · Edexcel IGCSE Mathematics A (4MA1) · Year 10

Question 1

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2015 · Question 1

1.	$50 \times 1.2 \times 120$ or 50×1.2 or 60 or 1.2×120 or 144			M1	Allow $\frac{50 \times 1.2}{120}$
		7200	2	A1	**Note:** SCB1 for 5000 or $\frac{50 \times 120}{1.2}$ oe or $41.6(66666) \times 120$ oe with 41.6(66666) rounded or truncated to at least 3SF
					Total 2 marks

Question 2

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 2

2	3/5 x 15 or 15 ÷ 5 × 3	9	2	M1 A1	M1 for 3/5 or 15 ÷ 5 × 3
					Total 2 marks

Question 3

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2019 · Question 1

1	73 ÷ 200 (=0.365) **or** 73 × 100 (= 7300) or 1 cm = 2 m oe "0.365" × 100 **or** "7300" ÷ 200 73 ÷ 2	36.5	3	M1 M1 A1	Allow their incorrectly converted 73 m ÷ 200	M2 for $100 \div \frac{200}{73}$ oe

Question 4

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2023 · Question 1

Q	Working	Answer	Mark	Notes	
1	$12 \div (5-2)$ (= 4) **or** 2 : 5 = 8 : 20 **or** A = 8 **or** S = 20 **or** eg $\frac{5}{15}x - \frac{2}{15}x = 12$ **or** $x = 60$		3	M1 for method to find the value of one share **or** working with the ratio for Arjun or Simon **or** setting up an equation **or** for finding the total number of goals (= 60)	M2 for $\frac{8}{5-2} \times 12$ oe
	eg $8 \times$ "4" **or** $8 \times \frac{8}{2}$ **or** 8 + 12 + 12 **or** $8 \times \frac{20}{5}$ **or** 20 + 12 **or** "60" $\times \frac{8}{15}$			M1 for a complete method	
	Correct answer scores full marks (unless from obvious incorrect working)	32		A1 SCB1 for $\frac{8}{15} \times 12 (= 6.4)$	
				Total 3 marks	

Question 5

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 1

1	345 ÷ 200 (=1.725) **or** 345 × 100(=34500)			M1 for a correct units conversion (×100) **or** ÷200
	"1.725" × 100 **or** "34500" ÷ 200		3	M1 for a correct units conversion (×100) **and** ÷200
		172.5		A1 accept 173 if at least M1 awarded
				Total 3 marks

Question 6

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2018 · Question 2

2	$\frac{3450}{2+6+7}$ (=230) **or** $\frac{2}{2+6+7}\times 3450(=460)$ **or** $\frac{7}{2+6+7}\times 3450(=1610)$ **or** $\frac{7-2}{2+6+7}$ $\left(=\frac{1}{3}\right)$		3	M1
	(7 – 2) × "230" **or** 7 × "230" – 2 × "230" **or** "1610" – "460" **or** "$\frac{1}{3}$"×3450			M1
		1150		A1
				Total 3 marks

Question 7

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2022 · Question 2

2	196 ÷ (9 – 5) (= 49) oe		3	M1	
	3 × "49"			M1	
		147		A1	SCB1 for an answer from 34.5 – 34.6 **or** an answer of 42
					Total 3 marks

Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2015 · Question 2

2	15 ÷ (6 − 4) (=7.5) "7.5" × 3 (=22.5)	22.5(0)	3	M1 M1dep A1 oe	or use of cancelled ratios (eg 3 : 6 : 4 = 0.75 : 1.5 : 1, 15÷0.5(=30)) or cancelled ratios, (eg 30×0.75) or M2 for $15 \div \frac{2}{3}$ oe
	Alternative				
	$\frac{6n}{13}-\frac{4n}{13}=15$ **or** $\frac{2n}{13}=15$ oe **or** $15\times\frac{13}{2}$ oe **or** n = 97.5		3	M1	
	$\frac{3}{3+6+4}$ × 97.5 oe			M1dep	
		22.5(0)		A1 oe	
	Alternative				
	(3 : 6 : 4).....21 : 42 : 28 and 24 : 48 : 32 22.5 : 45 : 30	22.5(0)	3	M1 M1dep A1 oe	For using ratios and seeing 21 : 42 : 28 and 24 : 48 : 32 Correct line **or** (21+24)÷2
					Total 3 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2017 · Question 3

3	Eg $\frac{715.5}{530}\times 750$ or 1.35×750 oe Or $\frac{750}{530}\times 715.5$ or $1.41(509...)\times 715.5$ oe Or $750 \div \frac{530}{715.5}$ or $715.5 \div \frac{530}{750}$ oe		3	M2	For a complete method If not M2 then M1 for $\frac{715.5}{530}$ or 1.35 oe or $\frac{530}{715.5}$ or 0.740(740…) oe or $\frac{750}{530}$ or 1.41(509...) oe or $\frac{530}{750}$ or 0.706(666…) oe or $530x = 750\times 715.5$
		1012.50		A1	Accept 1012.5
					Total 3 marks

Question 10

Edexcel IGCSE Mathematics A (4MA1) · Paper 2F · November 2023 · Question 15

Question	Working	Answer	Mark	Notes	
	SMALL (£ per g) $\frac{1.80}{1.50}(=0.012)$ or (p per g) $\frac{180}{150}(=1.2)$ or (g per £) $\frac{150}{1.80}(=83.(3...))$ or (g per p) $\frac{1.50}{1.80}(=0.83..)$ **LARGE** (£ per g) $\frac{5}{400}(=0.0125)$ or (p per g) $\frac{500}{400}(=1.25)$ (g per £) $\frac{400}{5}(=80)$ or (g per p) $\frac{400}{500}(=0.8)$ **COMPARISON OF SAME AMOUNTS EXAMPLE** eg cost of 1200 g = $\frac{1200}{150}\times 1.80(=14.4)$ or		3	M1 A start to comparing costs by finding cost per gram for either size bag or for finding grams per pound (or pence) for either size bag or for finding the cost of a weight of fudge for either size bag.	M2 for a fully correct method to reach a conclusion eg $\frac{400}{150}\times 1.80(=4.8)$ or $\frac{150}{400}\times 5(=1.875)$ or $\frac{5}{1.80}\times 150$ (=416.66…) or $\frac{1.80}{5}\times 400(=144)$ or 300 g = £3.60 100 g = £1.20 400 g = £4.80 etc
	cost of 1200 g = $\frac{1200}{400}\times 5(=15)$ oe				
				M1 2 correct compatible calculations	
	Working required	Small		A1 Correct conclusion with correct values from correct working dep on M2	
					Total 3 marks

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2018 · Question 4

4	675 ÷ (5 + 4) × 5 (= 375)	225	3	M1	M2 675 ÷ (5 + 4) × 3
	"375" ÷ 5 × 3			M1 dep M1	
				A1	
					Total 3 marks

Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2023 · Question 5

5	2 : 3 : 15 oe or 20 or (1 : 5) × 3 or (1 : 5 =) 3 : 15 or $2n$: $3n$: $15n$ e.g. 4 : 6 : 30 or G(reen) = 2, O(range) = 3 , Y(ellow) = 15		3	M1
	$\frac{2}{"20"}$ ′ 280 oe or 14 × 2 or $\frac{2}{"2"+"3"+"15"}$ ′ 280 oe or $\frac{2n}{"2n"+"3n"+"15n"}$ ′ 280 oe			M1
	Correct answer scores full marks (unless from obvious incorrect working)	28		A1 or 28 : 42 : 210 or 28 , 42 , 210 If not in this order must be labelled correctly
				Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2023 · Question 5

5	$390 \div (8-2)$ (= 65) or $\frac{8}{15}-\frac{2}{15}=390$ or $\frac{8}{15}x-\frac{2}{15}x=390$ or $\frac{6}{15}=390$ or $\frac{6}{15}x=390$ oe		3	M1	M2 for $\frac{390\times15}{6}$ oe
	"65" × (2 + 5 + 8) oe or $\frac{1}{15}=65$ or $\frac{1}{15}x=65$ or $\frac{1}{5}=195$ or $\frac{1}{5}x=195$			M1 or for 975 seen with further work and a different answer	
	Correct answer scores full marks (unless from obvious incorrect working)	975		A1 SCB1 for 52, 130, 208 or 390, 975, 1560 (or 2925) or 97.5, 243.75, 390 (or 731.25)	
					Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2019 · Question 7

7		55 ÷ (6 + 3 + 2) (= 5) (6 x "5") – (2 x "5")	20	3	M1 M1 A1	Or $\frac{6}{11}$ x 55 (= 30) or $\frac{2}{11}$ x 55 (= 10) Or M2 for Won = 30 and Lost = 10 (can be seen in a ratio 30 : 15: 10)
						Total 3 marks

Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2011 · Question 7

7.	3+5+7 or 15		3	M1	15 may be denominator of fraction or coefficient in an equation such as 15x = 90
	90 ÷ (3+5+7) or 90 ÷ "15" or 6 or $\frac{7}{15}$ oe			M1	dep
		42		A1	Also award for 18 : 30 : 42
					Total 3 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2015 · Question 7

7	$240\times\frac{3}{3+4+8}$ or 48 or $240\times\frac{8}{3+4+8}$ or 128		3	M1	M2 for $240\times\frac{5}{3+4+8}$
	"128" – "48"			M1 dep	
		80		A1	
					Total 3 marks

Question 17

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2017 · Question 7

7	96 ÷ 3 (= 32)		3	M1	M2 for $\frac{5}{3}\times96$
	9 × '32'(=288) **or** 4 × '32'(=128) **or** (9 − 4) × '32'			M1 dep	
		160		A1	
					Total 3 marks

Question 18

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2013 · Question 10

10	22 x 25000 (=550000) "550000" ÷100000	5.5	3	M1 or 25000 cm = 0.25km or 22 ÷100000 M1 or 22 x 0.25 or "0.00022" × 25000 A1
				Total 3 marks

Question 19

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2016 · Question 10

10	$\frac{24.5}{7} \times 2\ (=7)$ or $\frac{24.5}{7} \times 6\ (=21)$ or $\frac{24.5}{7} \times 8 (=28)$		3	M1
	$\frac{24.5}{7} \times 2 + \frac{24.5}{7} \times 6 + 24.5\ (7 + 21 + 24.5)$			M1 fully correct method [M2 for $\frac{24.5}{7} \times (7 + 6 + 2)$]
		52.5		A1
				Total 3 marks

Question 20

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · June 2019 · Question 11

11	15 – 3 : x – 3 = 2 : 7 **or** (15 – 3) ÷ 2 (= 6)	(n =) (15 – 3) ÷ $\frac{2}{2+7}$ (= 54) where n is the total age 3 years ago	3	M1	M2 for $\frac{(15-3)\times 7}{2}$ (= 42)
	$\frac{x-3}{15-3} = \frac{7}{2}$ oe **or** 7 × "6" (= 42)	"54" × $\frac{7}{2+7}$ (= 42)		M1	
		45		A1	
				Total 3 marks	

Question 21

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2013 · Question 11

11.	5 + 9 or 14 seen or or $\frac{n}{14}$ oe (provided no evidence of 14 from incorrect method)		3	M1 or $\frac{5+9}{5+9+6} \times x = 56$
	56 ÷ "14" or 4 or $\frac{6}{14} \times 56$			M1 dep or $56 \div \frac{14}{20}$ or 80
		24		A1 Also accept 20 : 36 : 24 as final answer
				Total 3 marks

Question 22

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2014 · Question 1

1	(a)	$900 \times \frac{13}{6}$		2	M1 for $\frac{900}{6}$ or 150 or $\frac{13}{6}$ (= 2.16…) oe or 900×13 or 11 700
			1950		A1 cao
	(b)	$6 \times \frac{1250}{750}$ or $1250 \div \frac{750}{6}$		2	M1 for $\frac{1250}{750}$ oe(= 1.66…) or $\frac{750}{1250}$ oe (= 0.6) or $\frac{750}{6}$ oe (=125)
			10		A1 cao
					Total 4 marks

Question 23

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · January 2016 · Question 1

1	a	eg. 24 ÷ 6 × 800			M1 for a complete method
			3200	2	A1 accept 3.2 litres
	b	eg. 450 ÷ 300 × 6			M1 for a complete method
			9	2	A1
					Total 4 marks

A* Maths Hub International Community Schools

Question 24

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2016 · Question 1

1	a	$60 \div 12 \times 150$ **or** $60 \div 12$ (=5) **or** $150 \div 12$ (=12.5)			M1	allow $x \div 12 \times 60$ oe where x is 300 **or** 250 **or** 100 **or** 2
			750	2	A1	
	b	$625 \div 250 \times 12$ oe			M1	complete method
			30	2	A1	
						Total 4 marks

Question 25

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2017 · Question 3

3	(a)	E.g. $\frac{300}{4} \times 10$			M1	for a correct scale factor or a correct first step E.g. $\frac{300}{4}$ **or** 75 **or** $\frac{10}{4}$ **or** 2.5 **or** $300 \div 4$ (=75)
			750	2	A1	
	(b)	E.g. $\frac{920}{115} \times 4$			M1	for a correct scale factor or a correct first step E.g. $\frac{920}{115}$ **or** 8 **or** $\frac{115}{4}$ **or** 28.75
			32	2	A1	
						Total 4 marks

Question 26

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2023 · Question 5

5	2 and 15 seen or 1×2 (+) 3×5 (= 17)	$2x + 15x$ (= 85) or $\frac{2}{3}y + 5y(=85)$ or $0.25t \times 2 + 0.75t \times 5$ (= 85)		4	M1	For 2 and 15 oe seen or 17 or a correct equation in one unknown for number of 2p coins (x) or number of 5p coins (y) or total number of coins (t)
	$85 \div (2 + 15)$ (= 5) or at least two pairs of multiples of the values of 2 and 15 (eg 4, 30; 6, 45…..) or 10(p) (and) 75(p) or 10 : 75 or 5×2 and 15×5 $2 \times 5 + 5 \times 3 \times 5$ or 20 coins	$17x = 85$ ($x = 5$) or $\frac{17}{3}y = 85$ ($y = 15$) or $4.25t = 85$ ($t = 20$)			M1	Assumes previous M1 for number of 2p coins **or** number of 5p coins **or** total number of coins **or** value of 2p coins **and** value of 5p coins may be **clearly** listed eg 2 555 2 555 2 555 2 555 2 555 with no ambiguity
	5 (2p coins) and 15 (5p coins) or 5 : 15 (if clearly identified (or used) as the key ratio eg not just part of a list) or $(3 - 1) \times 5$	eg $15 - 5$ oe			M1	Correct number of 2p coins **and** 5p coins or a sum to find the difference in number of coins
	Correct answer scores full marks (unless from obvious incorrect working)		10		A1	SCB1 if no other marks awarded for 21.25 in working or on answer line
						Total 4 marks

A+ Maths Hub International Community Schools

Question 27

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · November 2025 · Question 6

6	$\frac{275}{2+3}$ (= 55) **or** $\frac{275}{2+3}\times 2$ (= 110) **or** $\frac{275}{2+3}\times 3$ (= 165) **OR** $\frac{2}{5}\times\frac{3}{11}\left(=\frac{6}{55}\right)$ **or** $\frac{3}{5}\times 0.32\left(=\frac{24}{125}\right)$ oe		4	M1	for a method to find 1 part of the ratio **or** a method to find the amount of money that either Eli or Peta gets **OR** finds the proportion of the money that either Eli or Peta gives to charity Allow decimal equivalents eg 0.4 and 0.27(27…) throughout for M marks
	"110" × $\frac{3}{11}$ (= 30) **or** "165" × 0.32 (= 52.8(0)) **OR** "$\frac{6}{55}$" × 275 (= 30) **or** "$\frac{24}{125}$" × 275 (= 52.8(0)) **or** "$\frac{6}{55}$" + "$\frac{24}{125}$" $\left(=\frac{414}{1375}\right)$ **OR** $\left(1-\frac{3}{11}\right)\times$"110"(= 80) **and** (1 − 0.32) × "165" (= 112.2(0))			M1	for a method to find the amount of money that either Eli or Peta gives to charity **OR** finds total proportion of the money that will be given to charity **OR** for a method to find the amount of money that Eli keeps **and** the amount of money that Peta keeps
	"110" × $\frac{3}{11}$ (= 30) **and** "165" × 0.32 (= 52.8(0)) **OR** "$\frac{6}{55}$" × 275 (= 30) **and** "$\frac{24}{125}$" × 275 (= 52.8(0)) **OR** "$\frac{414}{1375}$" × 275 **OR** $\left(1-\frac{3}{11}\right)\times$"110" + (1 − 0.32) × "165" (= 192.2(0))			M1	for a method to find the amount of money that both Eli **and** Peta gives to charity **OR** for a complete method **OR** For a method to find the total amount of money both Eli and Peta keep
	Correct answer scores full marks (unless from obvious incorrect working)	82.8(0)		A1	
					Total 4 marks

Question 28

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2012 · Question 1

1.	(a)	$54\times\frac{5}{6}$		2	M1	for 54 × 5 or 270 or 54 ÷ 6 or 9 or $\frac{5}{6}$ oe or $\frac{6}{5}$ oe
			45		A1	cao
	(b)	ratio 36 : 5400 oe inc 0.36 : 54, 36 cm : 5400 cm, 0.36 m : 54 m (condone omission of units from one side) or fraction $\frac{5400}{36}$ oe inc $\frac{54\text{ m}}{0.36\text{ m}}$ (condone omission of units from either numerator or denominator)		3	M2	M1 for ratio or fraction with no units 0.36 or 3.6 or 36 or 360 or 3600... : 0.54 or 5.4 or 54 or 540 or 5400 ... oe $\frac{0.54 \text{ or } 5.4 \text{ or } 54 \text{ or } 540 \text{ or } 5400...}{0.36 \text{ or } 3.6 \text{ or } 36 \text{ or } 360 \text{ or } 3600...}$ oe eg 36 : 54, $\frac{54}{36}$, 36 : 540, $\frac{540}{36}$, 360 : 54, $\frac{54}{360}$, 1 : 1.5, 54 ÷ 36, 1 : 0.15
			150		A1	cao Do not award A1 for 150 cm, 150*n* etc
						Total 5 marks

Question 29

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2018 · Question 2

2	$170 \div 2$ (=85) **or** $170 \div 2 \times 7$ (=595) **or** $7 \div 2$ (=3.5) $7 \times$ "85" + 170 (=765) **or** $9 \times$ "85" (=765) **or** "595" + 170 (=765) **or** $170 \times$ "3.5" + 170 (=765) "765" ÷ 3 (=255) **or** "765" ÷ 3 × 5 (=1275) "255" × 2 **or** "1275" – "765" **or** "1275" ÷ 5 × 2	510	5	M1 M1 award of this mark implies the first M1 M1 dep on M2 M1 A1
	Alternative scheme			
2	(girls =) $\frac{2}{9}$ (of children) (girls =) $\frac{2}{9} \times \frac{3}{5} \left(= \frac{2}{15}\right)$ (of total) **or** G : C : A = $\frac{2}{9} \times \frac{3}{5} : \frac{3}{5} : \frac{2}{5} \left(= \frac{2}{3} : 3 : 2\right)$ "$\frac{15}{2}$" × 170 (=1275) **or** G : A = 2 : 6 oe "1275" ÷ 5 × 2 **or** 3 × 170	510	5	M1 M1 award of this mark implies the first M1 M1 dep on M2 M1 A1

Question 30

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · June 2017 · Question 3

3 (a)	250×97			M1	Completely correct method or figures 2425(0) e.g. 242.5
		24 250	2	A1	
(b)	4×500 (=2000) **or** $500 \div 93.5$ (=5.34759…)			M1	
	$(4 \times 500) \div 93.5$ **or** "5.34.." × 4			M1	
		21	3	A1	21 – 21.4
					Total 5 marks

Question 31

Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2012 · Question 6

6. (a)	1 + 7 or 8		2	M1	8 may be denominator of fraction or coefficient in an equation such as $8x = 32$	*SC* If M0 A0, award B1 for 4 : 28
		28		A1	cao	
(b)	32 × 45 or 1440 or 14.4(0)m		3	M1		
	$\frac{\text{"1440"}}{72}$			M1	dep	
		20		A1	cao	
					Total 5 marks	

Question 32

Edexcel IGCSE Mathematics A (4MA1) · Paper 3H · June 2015 · Question 7

7 (a)	450×1.16 oe		2	M1		
		522		A1		
(b)	850÷1.16 oe (= 732.76) **or** 732 – 733		3	M1	M1 for 3.50×1.16 (= 4.06)	
	"732.76" + 3.50			M1 (dep)	M1 (dep) for (850 + "4.06") ÷1.16 oe	
		736.26		A1	Accept 736 – 736.3	
					Total 5 marks	

CHALLENGE QUESTIONS

Question 33 *Edexcel IGCSE Mathematics A (4MA1) · Paper 4H · January 2014 · Question 1*

1. (a)	840 : 40 oe or 840 ÷ 40 oe or 1 : 21	21	2	M1 A1	Accept 21 : 1
(b)	(105 ÷ 3) x 2	70	2	M1 A1	M1 for 105 ÷ 3 (=35)
(c)	(105 ÷ {4+3}) x 3	45	2	M1 A1	M1 for 105 ÷ (4+3) (=15)
					Total 6 marks

Question 34 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2021 · Question 4*

4		eg (36 ÷ 9) × 5 **or** 20 [ducks] **or** 20 : 36 **or** for writing the 3 parts of the ratio correctly eg 35 : 10 : 18 oe		3	M1	For a fully correct calculation for the number of ducks or stating 20 ducks – may be shown in a ratio – does not need to be labelled if it is clear that the number or calculation refers to the number of ducks
		"20" ÷ 2 = 10 **and** 10 × 7 oe **or** $\frac{36}{18}\times 35$ oe			M1	For a correct calculation to find the number of chickens. (award the M2 for 70 : 20 : 36 or a different order if intention is clear eg by labels)
		Working not required, so correct answer scores full marks (unless from obvious incorrect working)	70		A1	
						Total 3 marks

Question 35 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2023 · Question 5*

Question	Working	Answer	Mark	Notes
5	eg (one share of the ratio =)120 ÷ 2 (= 60) **or** $120\times\frac{3}{2}$(=180) **or** 180 (g)[butter] **or** 180 : 120 **or** for writing the 3 parts of the ratio correctly eg 18 : 15 : 10 (or 18 : 15 and 15 : 10 or S : F = 18 : 10) oe eg 3.6 : 3 : 2		3	M1 For finding the value of one share or For a fully correct calculation for amount of butter or stating 180 (g) butter – may be shown in a ratio – does not need to be labelled if it is clear that the number or calculation refers to the amount of butter
	("60" × 3 ÷ 5) × 6 oe **or** $\frac{f}{\text{"180"}}=\frac{6}{5}$ **or** $\frac{18}{10}\times 120$ oe eg or $\frac{120}{10}\times 18$ oe or $\frac{3.6}{2}\times 120$ oe			M1 For a correct calculation to find the amount of flour Avril uses or a correct equation involving flour that if rearranged correctly would give the correct answer (award the M2 for 216 : 180 : 120 not labelled)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	216		A1 or flour = 216 or eg s = 120, b = 180, f = 216 (but flour must be clearly labelled)
				Total 3 marks

Question 36 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2020 · Question 4*

4	eg 76 ÷ (5 + 2 – 3) oe (= 19) or $5x + 2x - 3x = 76$ **and** $x = 76 \div (5 + 2 - 3)$ (=19) oe			M1 For a correct method to find the value of 1 share
	3 × "19" (= 57)			M1
	"57" – 48.5(0)			M1
		8.5(0)	4	A1
				Total 4 marks

Question 37

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · June 2025 · Question 6

Question	Working	Answer	Mark	Notes
6*	240 ÷ (3 + 4 + 5) (= 20) or $240\times\frac{3}{3+4+5}(=60)$ oe or $240\times\frac{4}{3+4+5}(=80)$ oe or $240\times\frac{5}{3+4+5}(=100)$ oe		4	M1 for a correct method to find the value of one share NB (240 ÷ 3 =) 80, (240 ÷ 4 =) 60 and (240 ÷ 5 =) 48 scores M0
	For two of (Pau:) 3 × "20" + 10 + 10 (= 80) or "60" + 10 + 10 (= 80) (Sam:) 4 × "20" – 10 (= 70) or "80" – 10 (= 70) (Tia:) 5 × "20" – 10 (= 90) or "100" – 10 (= 90)			M1 for the correct values for 2 of the people after S and T give P £10
	80, 70 and 90 or eg 9 : 7 : 8 oe or eg 4 : 3.5 : 4.5 oe			M1 for all 3 of 80, 70, and 90 correct (ignore units) or for the correct values for the final ratio in the wrong order (ignore units) or for the correct values for the final ratio unsimplified (ignore units)
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	8 : 7 : 9		A1 other orders are acceptable if labelled correctly
				Total 4 marks

A+ Maths Hub International Community School

Question 38 *Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · January 2020 · Question 12*

12	**Litres per amount of money and then conversion**				
	$\frac{8.6\times10^5}{770000}(=1.1168)$ *l*/$				M1 Number of litres per $ for D
	$\frac{4.2\times10^5}{2500000}(=0.168)$ *l*/k				M1 Number of litres per Krone for A
	A: *l*/$ to *l*/k '1.1168' ÷ 6.57 (= 0.1699..)or D: *l*/k to *l*/$ '0.168' × 6.57 (= 1.103..)				M1 *l*/$ to *l*/k for A or *l*/k to *l*/$ for D
			Arctic Oil and relevant figures		A1 for Arctic Oil with 1.1168… and 1.10376… **or** 0.168 and 0.1699..
	Conversion then litres per amount of money				
	$\frac{2500000}{6.57}$ (=380517.5..) or 770 000 × 6.57(= 505 8900)				M1 Changing Krone to $ or $ to Krone
	$\frac{4.2\times10^5}{2500000}(=0.168)$ or $\frac{4.2\times10^5}{'380517.5'}(=1.103..)$				M1 Litres per Krone or litres per $ for D
	$\frac{8.6\times10^5}{770000}(=1.1168)$ or $\frac{8.6\times10^5}{'5058900'}(=0.1699..)$				M1 Litres per Krone or litres per $ for A
			Arctic Oil and relevant figures		A1 for Arctic Oil with 1.1168… and 1.10376… **or** 0.168 and 0.1699..
	Cost per litre then conversion				
	$\frac{2500000}{4.2\times10^5}(=5.952..)$				M1 Price per litre in Krone for D
	$\frac{770000}{8.6\times10^5}(0.895..)$				M1 Price per litre in $ for A
	'5.952' ÷ 6.57(=0.9059..) or '0.895' × 6.57(= 5.882..)				M1 Conversion of Krone to $ or $ to Krone
			Arctic Oil and relevant figures		A1 For Arctic Oil with 5.952 and 5.882 **or** 0.895 and 0.9059
	Conversion then cost per litre				
	$\frac{2500000}{6.57}$ (=380517.5..) or 770 000 × 6.57(= 505 8900)				M1 Changing Krone to $ or $ to Krone
	$\frac{2500000}{4.2\times10^5}(=5.952)$ or $\frac{'380517.5..'}{4.2\times10^5}(=0.9059..)$				M1 Cost per litre in Krone or cost per litre in $ for D
	$\frac{770000}{8.6\times10^5}(=0.895)$ or $\frac{'5058900'}{8.6\times10^5}(=5.882..)$				M1 Cost per litre in $ or cost per litre in Krone for A
			Arctic Oil and relevant figures		A1 For Arctic Oil with 5.952 and 5.882 **or** 0.895 and 0.9059
	Comparing equal amounts				
	$\frac{8.6\times10^5}{4.2\times10^5}(=\frac{43}{21}=2.047..)$	$\frac{4.2\times10^5}{8.6\times10^5}(=\frac{21}{43}=0.488..)$			M1 Multiplier for same amount of D as A or same amount of A as D
	'2.047..'×2500 000 K (=5119047.619..)K	'2.047..' × 770 000 $ (=376046.511..)$			M1 Cost of equal amount of D as A or A as D
	'5119047.619'÷6.57 = 779154.88…$ or 770 000×6.57=5058900 K	'376046.511..'× 6.57 =2470625.58..K or 2500 000÷6.57 = 380517..$			M1 Converts so can compare costs – either K to $ or original A to K or $ to K or original D to $
			Arctic Oil and relevant figures		A1 Arctic Oil and 779154.. or with 2470625..(figures may be rounded) Or Arctic Oil with 5119047… and 5058900 or with 376046.. and 380517
Students may compare other equal amounts – please use the scheme that best fits their method and award marks appropriately.					
					Total 4 marks